



Syllabus: ENBC311 – Python for Data Analysis

Instructor

Dr. Bahaa Ghammraoui (bahaag@umd.edu)

Office Hours: Online: Fridays, 3 to 4 pm (Available most days/times by e-mail or by appointment)

Location: TBD

About the Course

Time: M 5:00pm - 7:30pm (BLD3 4206 and Online - Zoom); Online: Course ELMS Site - Discussion, Assignments/Projects and Exams.

Location: Online through Zoom, Synchronous, and Recorded (including office hours, AJC TBD); Online: <http://elms.umd.edu>

Attendance is highly encouraged and there will be bonus points provided based on participation.

All sessions will be recorded and made available to students who miss class.

Description

Aims to provide an introduction to structured programming, computational methods, and data analysis techniques, with the goal of equipping students with a foundation that allows them to confidently tackle problems in research and industry. The course begins with basic programming concepts such as data types, operators, logical flow control, debugging, and computational complexity using Python. It then delves into more advanced topics such as dynamic programming, algorithms, and object-oriented programming. Students will also be introduced to effective tools and libraries for data analysis and visualization. Towards the end of the course, there will be an overview of machine learning, artificial intelligence, neural networks, and convolution neural networks, along with some useful libraries and modules.

At the end of the course, students will:

- Have a solid understanding of basic programming concepts, such as variables, functions, etc.
- Understand the process of creating algorithms, writing programs, debugging code.
- Confidently write programs to solve many common problems that arise in their studies and work.
- Be able to apply Python for data management, data exploration, and data analysis.
- Have a solid foundation to build upon should they desire to study more advanced programming.

Prerequisites

While this class assumes students have little or no programming experience (particularly with Python), it is still useful to have some familiarity with basic programming and logical things.

The following courses are prerequisite to taking the class:

Calculus I and II

Linear Algebra

Textbooks (No Required Textbooks)

Handouts and summary for each chapter will be provided.

Online References (Highly Recommended)



Official Python 3 Documentation - [The Python Tutorial — Python 3.12.4 documentation](#)

Python Tutorials – For example:

<https://www.geeksforgeeks.org/python-programming-language-tutorial/>

<https://www.w3schools.com/python/>

Grading

Students are encouraged to attend and actively participate in each lecture and the online discussion board. Assignments will be given on a weekly basis to help students gain practical experience in developing programs with Python. Final grades will be determined by the following items:

- Assignments = 30
- Mid-term Exam 1 = 20%
- Mid-term Exam 2 = 20%
- Final Exam with Final Project = 30%

Grades will follow the standard scale: A+. 97-100. 4.0. A. 93-96. 4.0. A-. 90-92. 3.7. B+. 87-89. 3.3. B. 83-86. 3.0. B-. 80-82. 2.7. C+. 77-79.; D+. 68.75. D. 65.00. D-. 61.25. E<60.

Curving is at the discretion of the professor.

Assignments

I will assign homework roughly every week; most problems. These problems will serve as your study guide for exams. Solutions for these will be provided after the fact.

A late assignment may incur a penalty (approved excuse will be reduced by 30%), depending on the extent and circumstances. I will drop your lowest assignment score before computing your average.

Collaborate!!! I encourage everybody to collaborate in the assignments, but must note on their assignments who they collaborated with as a comment at the top of the file

Mid-term and Final Projects:

- You can think of these as larger assignments to be completed on your own (no collaboration).

Software

A personal computer with Python 3 (<https://www.python.org/downloads/>) is required to complete the assignments for this course. While students may use any Python 3 IDE, the only supported environment for this course is an up to date installation of Anaconda ([Download Anaconda Distribution | Anaconda](#)). We will use Jupyter Notebook for demonstration.



Communication

Email

The instructor will always be available by email (bahaag@umd.edu) . You can also use the messenger in ELMS to send an email to me at any time. I will try to answer within 24 hours and usually sooner.

Online / Chatroom Office Hours

As needed, I will provide online office hours. I will do this Fridays from 5 to 6 pm, but you can also send me an e-mail to request a time to meet. We will use Zoom for us to meet remotely and all these meetings will also be recorded and available for those who couldn't attend.

In person: You can also email me to set up individual office hours by appointment.

Class Attendance and Environment

You are strongly recommended to attend every lecture, since much of the material in this course will only be covered in lecture. This class will also use a blended learning system. Students will meet in a virtual space online (ELMS) which will be treated as a classroom. It is important to recognize that the classroom is an environment that requires respect for all participants. Students are expected to conduct themselves in a considerate manner both in-person as well as online.

Disabilities and Religion

Any student with a disability is encouraged to meet with the instructor privately during the first week of class to discuss accommodations. I will make every effort to accommodate students who are registered with the Disability Support Services (DSS) Office and provide a DSS accommodation form. Please refer to the Online Undergraduate Catalog Policy on Religious Observance.

Academic Integrity

The University of Maryland, College Park has a nationally recognized Code of Academic Integrity, administered by the Student Honor Council. This Code sets the standards for academic integrity at Maryland for all undergraduate and graduate students. As a student, you are responsible for upholding these standards for this course. It is very important for you to be aware of the consequences of cheating, fabrication, facilitation, and plagiarism. For more information on the Code of Academic Integrity or the Student Honor Council, please visit <http://www.shc.umd.edu>.



Course Schedule (Subject to Change and the final schedule will be made available one week before the class start)

Class#	Date	Topic			Homework #
1		Introduction to Python And what is algorithm?			
2		Python Data Types			
3		Basic Input-Output programming and Mathematical Functions in Python			HW1
4		Looping Techniques (1)			
5		Looping Techniques (2)			
6		Functions and Modules			HW2
7		File processing: reading and writing (1)			
8		File processing: reading and writing (2)			HW3
9		Midterm review # 1			
10		Midterm #1			
11		Object-Oriented Programming (OOP) in Python (1)			
12		Object-Oriented Programming (OOP) in Python (2)			HW4
13		Object-Oriented Programming (OOP) in Python (3)			
14		NumPy Arrays and linear algebra (1)			
15		NumPy Arrays and linear algebra (2)			
16		Visualization libraries in Python			HW5
17		Data analysis and Visualization with			



		Pandas module (1)			
18		Data analysis and Visualization with Pandas module (2)			HW6
19		Making predications by interpolation and extrapolation			HW7
20		Midterm review #2			
21		Midterm # 2			
22		Numerical Methods for Root Finding			
23		Numerical Differentiation and Integration			HW8
24		Machine learning/Neural Networks - Create a Simple Neural Network in Python			
25		Convolutional Neural Networks - Create a Convolutional Neural Network in Python			HW9
26		Machine/deep learning with TensorFlow (1)			
27		Machine/deep learning with TensorFlow (2)			HW10
28		Final Exam review 1			
29		Final Exam review 2			
30		Final Exam			